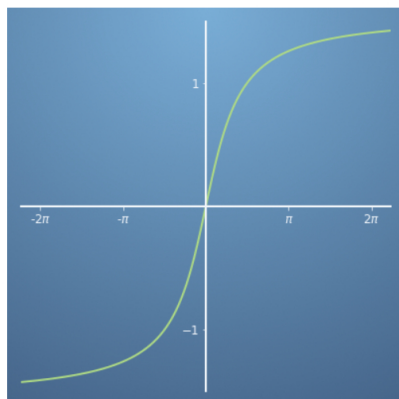


1. The graph below shows the function $f(x) = \tan^{-1}(x)$

1 point

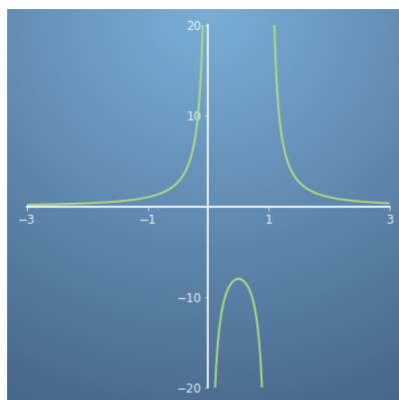


By using the Maclaurin series or otherwise, determine whether the function shown above is even, odd or neither.

- ☒ Odd
☐ Even
☐ Neither odd nor even

2. The graph below shows the discontinuous function $f(x) = \frac{2}{(x^2-x)}$. For this function, select the starting points that will allow a Taylor approximation to be made.

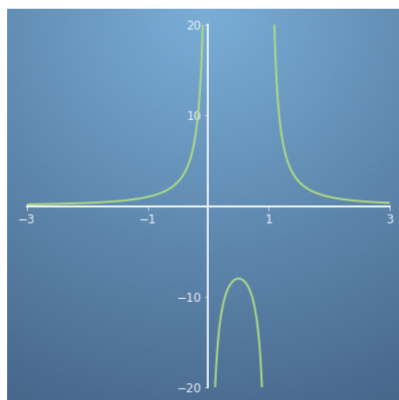
1 point



- ☐ $x = 1$
☒ $x = -3$
☒ $x = 0.5$
☒ $x = 2$

3. For the same function as previously discussed, $f(x) = \frac{2}{(x^2-x)}$, select all of the statements that are true about the resulting Taylor approximation.

1 point

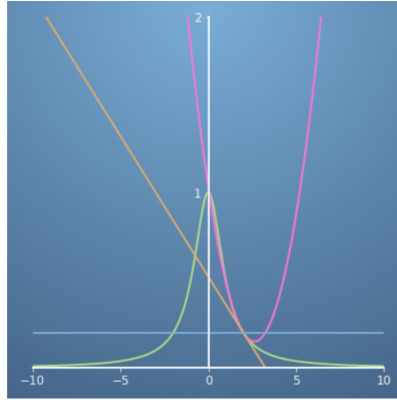


- ☐ The approximation converges quickly
- ☒ Approximation ignores segments of the function
- ☐ This is a well behaved function
- ☒ Approximation ignores the asymptotes
- ☐ Approximation accurately captures the asymptotes

4. The graph below highlights the function $f(x) = \frac{1}{(1+x^2)}$ (green line), with the Taylor expansions for the first 3 terms also shown about the point $x = 2$. The Taylor expansion is

1 point

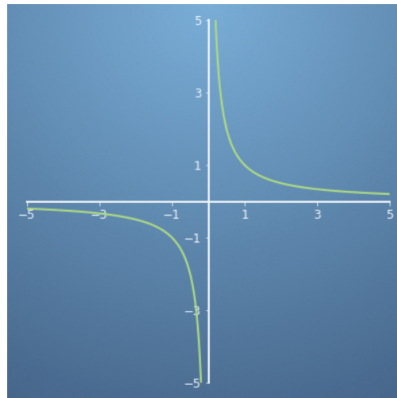
$f(x) = \frac{1}{5} - \frac{4(x-2)}{25} + \frac{11(x-2)^2}{125} + \dots$ Although the function looks rather normal, we find that the Taylor series does a bad approximation further from its starting point, not capturing the turning point. What could be the reason why this approximation is poor for the function described.



- ☐ The function has no real roots
- ☐ None of these options
- ☐ Function does not differentiate well
- ☒ Asymptotes are in the complex plane
- ☒ It is a discontinuous function in the complex plane

5. For the function $f(x) = \frac{1}{x}$, provide the linear approximation about the point $x = 4$, ensuring it is second order accurate.

1 point



- ☐ $f(x) = 1/4 - x/16 + O(\Delta x^2)$
- ☐ $f(x) = 1/4 + x/16 - O(\Delta x^2)$
- ☐ $f(x) = 1/4 - (x-4)/16 + O(\Delta x)$
- ☒ $f(x) = 1/4 - (x-4)/16 + O(\Delta x^2)$